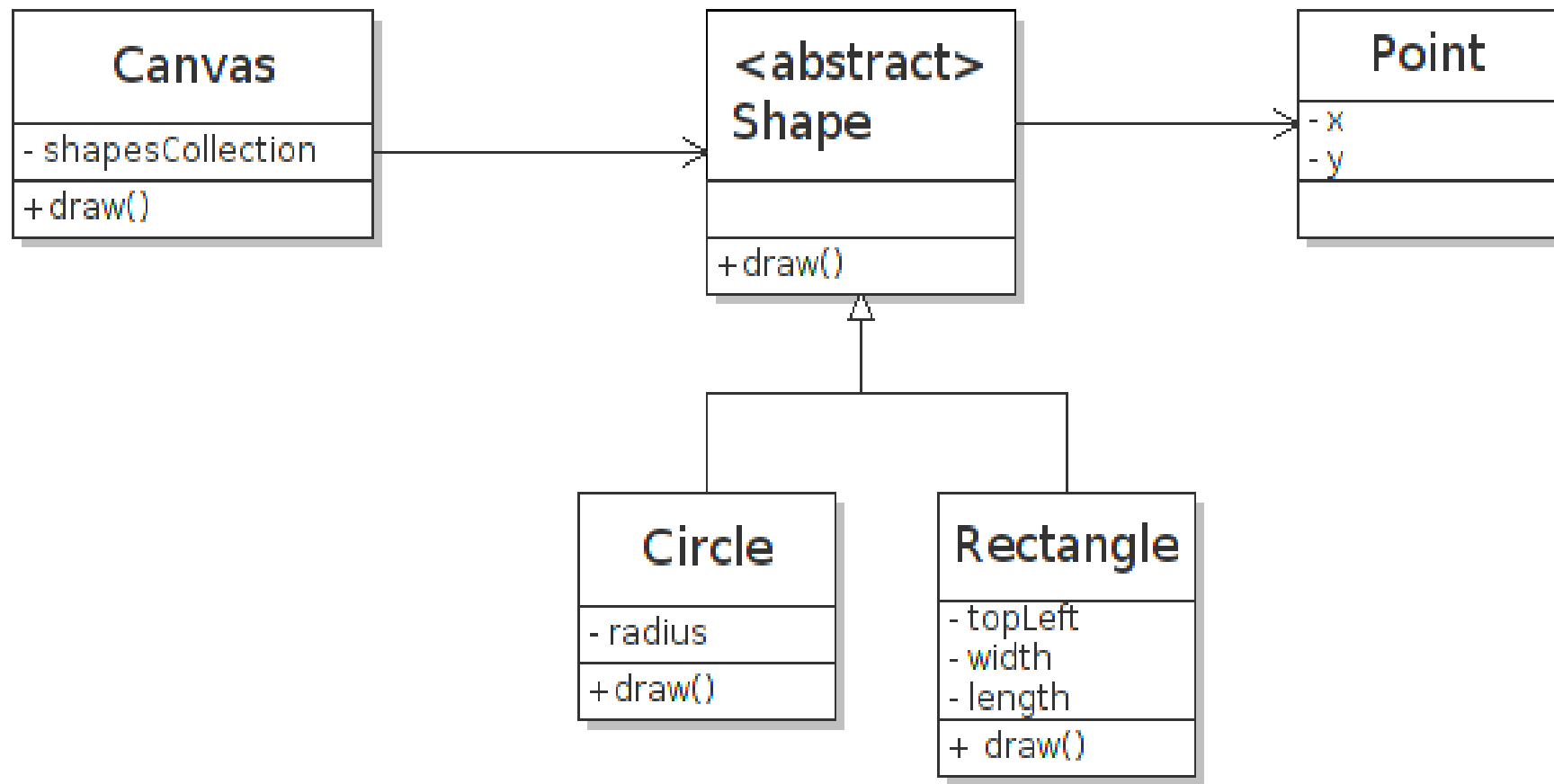


Serialization

- Process of writing the state of an object to an output stream (a serial sequence of bytes / characters)
- Generally used in context of writing to a more permanent form rather than transient (in memory) data
- Common uses include:
 - Storage of objects to file systems
 - Transportation of objects in case of distributed systems
- Serialized objects can be deserialized to reconstruct the object in memory

Graphics Example



```
class Point {  
    private int x;  
    private int y  
  
    public Point(int x,int y) {...}  
    public int getX() {...}  
    public int getY(){ ... }  
}
```

Point p1 = new Point(100,150);

int x1 = p1.getX();

int y1 = p1.getY();

0x100

0 .. 3

100

4 .. 7

150

Point p2 = new Point(200,250);

int x2 = p2.getX();

int y2 = p2.getY();

0x200

0 .. 3

200

4 .. 7

250

```

abstract class Shape {
    private Point point;

    public Shape(int x,int y) {
        point = new Point (x,y);
    }

    public abstract void draw();
}

```

```

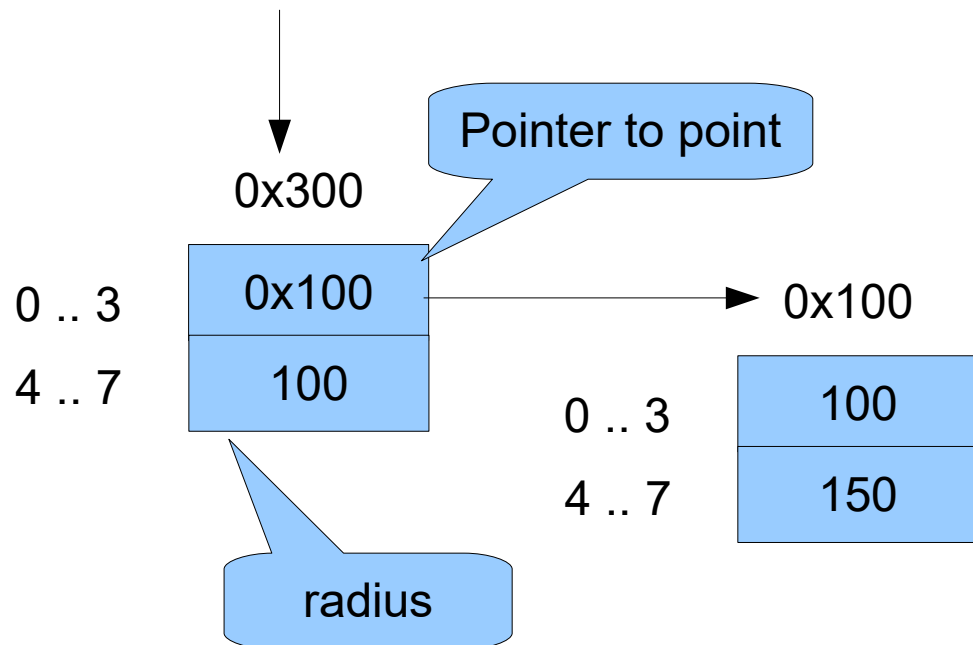
class Circle extends Shape {
    private int radius;

    public Circle (int x,int y,int r) {
        super(x,y) {
            radius = r ;
        }

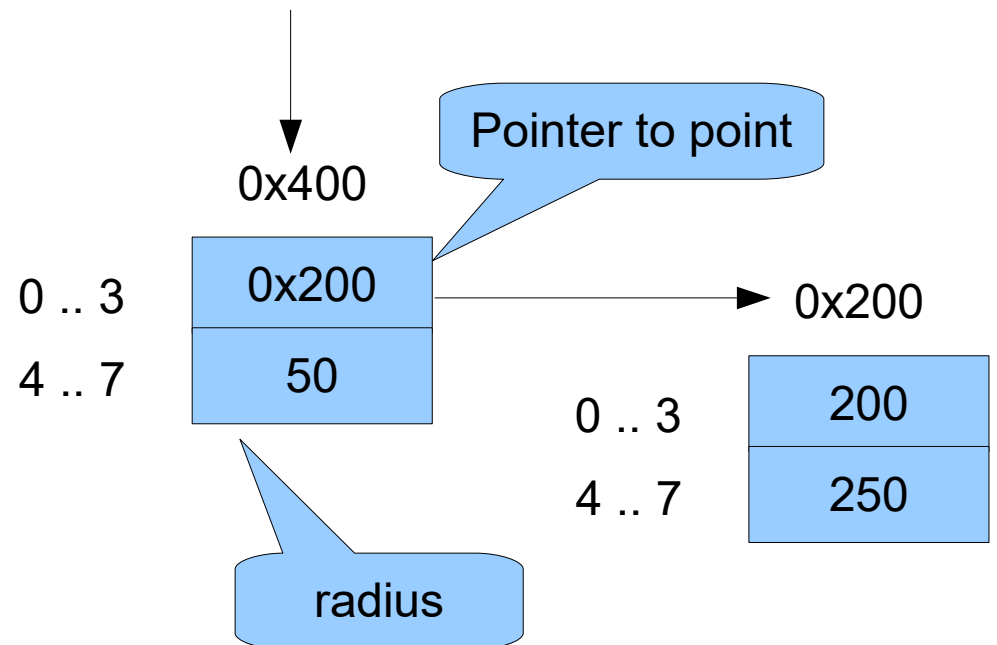
    public void draw() { ... }
}

```

Shape c1 = new Circle(100,150,100);



Shape c2 = new Circle(200,250,50);



Serialized Examples

```
Shape c1 = new Circle(100,150,100);
```

0x300

Pointer to point

0x100

0x100

100

0 .. 3

100

4 .. 7

150

radius

0 .. 3

100

x

4 .. 7

150

y

8 .. 11

100

radius

Without pointer

0 .. 3

8

pointer

4 .. 7

100

radius

8 .. 11

100

x

12 .. 15

150

y

With pointer

Serialization Formats

- Binary

- More compact and efficient
- Generally architecture dependent
- Not human readable, consequently secure from tampering
- Byte ordering / Memory layout based as well as independent

- Text-based

- More bloated but compression techniques can be applied
- Mostly architecture independent
- Human readable, more flexible but prone to tampering
- Examples : XML, JSON, etc.

Serialization using XML

Shape c1 = new Circle(100,150,100);

0x300

Pointer to point

0 .. 3
4 .. 7

0x100
100

0x100

0 .. 3

100

4 .. 7

150

radius

```
<stream>
  <circle>
    <point>
      <x>100</x>
      <y>150</y>
    </point>
    <radius>100</radius>
  </circle>
</stream>
```

Without pointer

```
<stream>
  <circle>
    <point type="pointer">
      1
    </point>
    <radius>100</radius>
  </circle>
  <point id="1">
    <x>100</x>
    <y>150</y>
  </point>
</stream>
```

With pointer

Serialization using JSON

```
Shape c1 = new Circle(100,150,100);
```

0x300

Pointer to point

0x100

0x100

100

0 .. 3

100

4 .. 7

150

radius

```
{  
  type: circle,  
  point : {  
    type: Point,  
    x: 100,  
    y: 150  
  }  
  radius: 100  
}
```

Without pointer

```
[ {  
  type: circle,  
  radius : 100.  
  point : { type : pointer, value : 1}  
},  
{  
  id : 1,  
  type : Point,  
  x : 100,  
  y : 150  
}]
```

With pointer